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NIXON & VANDERHYE, PC
901 NORTH GLEBE ROAD, 11TH FLOOR
ARLINGTON, VA 22203

EXAMINER	
DIXON, ANNETTE FREDRICKA	

ART UNIT	PAPER NUMBER
3785	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

PTOMAIL@nixonvan.com
pair_nixon@firsttofile.com

Office Action Summary	Application No. 17/426,362	Applicant(s) BARLOW et al.	
	Examiner ANNETTE F DIXON	Art Unit 3785	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 7/28/2021.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-26 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-26 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 7/28/2021 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date ALL IDS.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

1. Primary Examiner acknowledges Claims 1-26 are pending this application, with Claims 1, 3-10, 12, 14-16, and 23-26 having been currently amended, Claims 27-95 having been cancelled by preliminary amendment on July 28, 2021.

Notice of Pre-AIA or AIA Status

2. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claim Rejections - 35 USC § 103

3. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

4. Claims 1-21 and 23-25 are rejected under 35 U.S.C. 103 as being unpatentable over Ovzinsky et al. (9,731,090) in view of Veliss et al. (2008/0060649).

As to Claim 1, Ovzinsky discloses a positioning and stabilizing structure (Figure 6) to provide a force to hold a seal forming structure (510 - "the patient interface 510 (e.g., nasal prong/nozzle arrangement)" Column 7, Lines 30-35) in a therapeutically effective position on a patient's head for positive pressure airway treatment of sleeping disordered breathing ("The present technology relates to air delivery conduits used in Positive Airway Pressure (PAP)

systems for treatment, e.g., of Sleep Disordered Breathing (SDB)" Column 1, Lines 10-20), the positioning and stabilizing structure comprising: at least one gas delivery tube (x20, "each conduit 520 may be adapted to engage a respective end or inlet of the patient interface 510 (e.g., nasal prong/nozzle arrangement) … each conduit 520 is adapted to extend from adjacent to or under the patient's nose, over the patient's cheeks, between the patient's eye and ear, and terminate at the crown of the patient's head." Column 7, Lines 30-40) to deliver a flow of air to an entrance of the patient's airways via a seal forming structure (510), each gas delivery tube (x20) being constructed and arranged to contact, in use, at least a region of the patient's head superior to an otobasion superior of the patient's head ("each conduit 520 is adapted to extend from adjacent to or under the patient's nose, over the patient's cheeks, between the patient's eye and ear, and terminate at the crown of the patient's head." Column 7, Lines 30-40), each gas delivery tube (x20) comprising a tube wall (best seen Figure 10) defining a hollow interior through which air is able to flow to the seal forming structure (510), each gas delivery tube (x20) comprising: a superior tube portion (x20(2) - "the second cuff or end 520(2) may be adapted to engage a respective end of a manifold 515 communicated with the outlet of the PAP device via another air delivery conduit" Column 7, Lines 35-40) configured to extend in use from a position on top of the patient's head to a position along the side of the patient's head or face; and an inferior tube portion (x20(1) - "a first cuff or end 520(1) of each conduit 520 may be adapted to engage a respective end or inlet of the patient interface 510 (e.g., nasal prong/nozzle arrangement)" Column 7, Lines 30-35) configured to extend from the position along the side of the patient's head or face to a position adjacent the entrance to the patient's airways.

Regarding the composition of the gas delivery tube (x20) of each of the superior tube portion and inferior tube portions, as best seen in Figures 10-14, Ovzinsky discloses "an air delivery conduit 620 according to another example of the disclosed technology. In this example, the air delivery conduit 620 includes first and second conduit portions 621, 622 that cooperate to form the conduit, i.e., first conduit portion provides a portion of the conduit circumference and the second conduit portion provides the remaining portion of the conduit circumference" (Column 12, Lines 30-45), wherein "Each conduit portion 621, 622 includes an inner layer of a film laminate 650 that forms an interior surface of the conduit and an outer layer of a textile or fabric 655 that forms an exterior surface of the conduit. In an example, the film laminate is a polyurethane or medical grade film, and the textile is a thermoformable fabric" (Column 12, Lines 45-50). Thus, Ovzinsky clearly discloses the claimed orientation whereby an air impermeable material ("an inner layer of a film laminate 650") is provided to the textile material ("an outer layer of a textile or fabric 655") to make the tube portion air impermeable.

Yet, Ovzinsky **does not expressly disclose** "one of the inferior tube portion and superior tube portion has a greater rigidity than the other of the inferior tube portion and superior tube portion".

Veliss teaches a similar positioning and stabilizing structure (Figures 1-1 thru 3-3) to that of Ovzinsky having a gas delivery tube (42, "the tubing 40 includes two tubes or inlet conduits 42 (also referred to as gas passages or gas conduits) communicated with the interfacing structure 20 to deliver breathable gas to the interfacing structure 20 (e.g., see FIG. 1-6)." Para 0102) for delivering a flow of air to an entrance of the patient's airways via a seal forming structure (20, "the tubing 40 includes two tubes or inlet conduits 42 (also referred to as gas

passages or gas conduits) communicated with the interfacing structure 20 to deliver breathable gas to the interfacing structure 20 (e.g., see FIG. 1-6)." Para 0102) having a superior tube portion (42.2, best seen Figure 2-2) and an inferior tube portion (42.1, best seen Figure 2-1).

Regarding the remaining limitations of the claims, Veliss teaches the superior tube portion (42.2, best seen Figure 2-2) has a thicker construction - which equates to more stiffness or rigidity; whilst, the inferior tube portion (42.1, best seen Figure 2-1) has a thinner construction - which equates to less stiffness or rigidity. Veliss teaches the use of a "rigidizing element 50 may have a varying thickness along its length, e.g., to vary the stiffness or rigidity of the tube 42 along its length" (Para 0131), wherein the thicker regions provide "more shape-holding, self-supporting, and/or robust" support (Para 0133); whilst, the thinner regions "may be adjusted for comfort" (Para 0133).

Therefore, it would have been obvious to modify the superior and inferior tube portions of the gas delivery tube of Ovzinsky to have a thicker region at the superior tube portion and a thinner region at the inferior tube portion whereby the thickness and in turn stiffness/rigidity of the gas delivery tube is varied along its length as taught by Veliss for the purpose of providing more rigidity at the superior tube portion to impart better shape holding, self-supporting and a robust construction for the connection to the gas pressure supply tube while the providing less rigidity at the interior tube portion to impart better comfort for the patient at the connections proximate to the face of the patient.

As to Claim 2, the modified Ovzinsky, specifically Ovzisky discloses a superior tube portion (x20(2) - "the second cuff or end 520(2) may be adapted to engage a respective end of a manifold 515 communicated with the outlet of the PAP device via another air delivery conduit"

Column 7, Lines 35-40); yet, **does not expressly disclose** "a first end configured to lie against a superiorly-facing surface of the patient's head in use and a second end configured to lie against a laterally-facing surface of the patient's head in use".

Veliss teaches the superior tube portion (42.2, best seen Figure 2-2) commensurate in scope with the superior tube portion (x20(2)) of Ovzisky. Regarding the first end (44, "the tube 42 has a generally D-shaped cross-section and includes an internal or inwardly facing surface 44 and an external or outwardly facing surface 45." Para 0120, also see Para 0121) of the superior tube portion (42.2, best seen Figure 2-2) having a flattened orientation (Figures 3-1 thru 3-4f; and best seen Figure 4-1 on the patient) for lying adjacent the patient's head in use ("The internal surface 44 is relatively flat and adapted to sit substantially flush against the patient's face in use. The internal surface 44 may have a tapered configuration form an inner edge to an outer edge to provide a comfortable fit for a wide range of patients. The internal surface 44 provides a relatively large surface area which results in a more even load distribution. This arrangement is less likely to create pressure points in use. Also, the internal surface 44 may have grip-like material to help stabilize the patient interface on the patient's face." Para 0121), and the second end (45, "the tube 42 has a generally D-shaped cross-section and includes an internal or inwardly facing surface 44 and an external or outwardly facing surface 45." Para 0120, also see Para 0122) of the superior tube portion (42.2, best seen Figure 2-2) having a curved orientation (Figures 3-1 thru 3-4f; and best seen Figure 4-1 on the patient) for lying laterally against the patient's head in use ("The external surface 45 has a smooth contour that blends with the patient's face. That is, the external surface 45 has a profile or organic form with edges that blend into the patient's face, e.g., in a tangential manner, to prevent any edges from

catching on bedclothes, pillows, etc., during sleep (e.g., when the patient rolls over)." Para 0122).

Therefore, it would have been obvious to further modify the superior tube portion of the modified Ovzinsky to include the first end and the second end as taught by Veliss to provide a "D shaped cross section" in which the first end provides "more even load distribution", "less likely to create pressure points in use", and "stabilize the patient interface on the patient's face" while the second end "prevent any edges from catching on bedclothes, pillows, etc., during sleep (e.g., when the patient rolls over)".

As to Claim 3, the modified Ovzinsky, specifically Ovzisky discloses an inferior tube portion (x20(1) - "a first cuff or end 520(1) of each conduit 520 may be adapted to engage a respective end or inlet of the patient interface 510 (e.g., nasal prong/nozzle arrangement)" Column 7, Lines 30-35); yet, **does not expressly disclose** "a first end configured to lie against a laterally-facing surface of the patient's head in use and a second end configured to connect with the seal-forming structure."

Veliss teaches the inferior tube portion (42.1, best seen Figure 2-1) commensurate in scope with the inferior tube portion (x20(1)) of Ovzisky. Regarding the first end (44, "the tube 42 has a generally D-shaped cross-section and includes an internal or inwardly facing surface 44 and an external or outwardly facing surface 45." Para 0120, also see Para 0121) of the inferior tube portion (42.1, best seen Figure 2-1) having a flattened orientation (Figures 3-1 thru 3-4f; and best seen Figure 4-1 on the patient) for lying adjacent the patient's head in use ("The internal surface 44 is relatively flat and adapted to sit substantially flush against the patient's face in use. The internal surface 44 may have a tapered configuration form an inner edge to an

outer edge to provide a comfortable fit for a wide range of patients. The internal surface 44 provides a relatively large surface area which results in a more even load distribution. This arrangement is less likely to create pressure points in use. Also, the internal surface 44 may have grip-like material to help stabilize the patient interface on the patient's face." Para 0121), and the second end (45, "the tube 42 has a generally D-shaped cross-section and includes an internal or inwardly facing surface 44 and an external or outwardly facing surface 45." Para 0120, also see Para 0122) of the inferior tube portion (42.1, best seen Figure 2-1) having a curved orientation (Figures 3-1 thru 3-4f; and best seen Figure 4-1 on the patient) for lying laterally against the patient's head in use ("The external surface 45 has a smooth contour that blends with the patient's face. That is, the external surface 45 has a profile or organic form with edges that blend into the patient's face, e.g., in a tangential manner, to prevent any edges from catching on bedclothes, pillows, etc., during sleep (e.g., when the patient rolls over)." Para 0122). As best seen in Figure 2-1, the second end (45) of the inferior tube portion (42.1, best seen Figure 2-2) will engage with the seal forming structure (20).

Therefore, it would have been obvious to further modify the inferior tube portion of the modified Ovzinsky to include the first end and the second end as taught by Veliss to provide a "D shaped cross section" in which the first end provides "more even load distribution", "less likely to create pressure points in use", and "stabilize the patient interface on the patient's face" while the second end "prevent any edges from catching on bedclothes, pillows, etc., during sleep (e.g., when the patient rolls over)" such that the second end can engage with the seal forming structure.

As to Claim 4, the modified Ovzinsky, specifically Ovzisky discloses the inferior tube portion (x20(1)) fluidly connects the superior tube portion (x20(2)) and the seal forming structure (510) as an intermediary structure best seen Figure 6.

As to Claim 5, the modified Ovzinsky, specifically Ovzisky discloses the superior tube portion (x20(2)) is adapted to receive the flow of air from a source via a crown connector (515, best seen Figure 6, "the second cuff or end 520(2) may be adapted to engage a respective end of a manifold 515 communicated with the outlet of the PAP device via another air delivery conduit." Column 7, Lines 30-40).

As to Claim 6, please see the rejection of Claims 1 and 5, wherein Claim 5 recites the feature of the crown connector (515) and Claim 1 discloses the "at least one gas delivery tube". The difference lies in the fact that the "at least one gas delivery tube" now has a "pair of gas delivery tubes". The modified Ovzinsky, specifically Ovzisky discloses the gas delivery tube extending on both sides of the head and thus making a "pair of gas delivery tubes".

As to Claim 7, the modified Ovzinsky, specifically Ovzisky discloses the construction of the gas delivery tube (x20) having dual layers (best seen Figures 10-14) wherein "The film laminate 650 is applied to the fabric 655 (FIG. 11), and then the fabric and laminate are both thermoformed to create symmetrical complementary shapes (FIG. 12), i.e., conduit portions 621, 622." (Column 12, Lines 45-55); yet, **does not expressly disclose** "the first layer of material is more rigid than the second layer of material".

Veliss teaches the use of a "rigidizing element 50 may have a varying thickness along its length, e.g., to vary the stiffness or rigidity of the tube 42 along its length" (Para 0131), wherein the thicker regions provide "more shape-holding, self-supporting, and/or robust" support (Para

0133); whilst, the thinner regions "may be adjusted for comfort" (Para 0133). Veliss teaches the application of the first layer of material (50) as more rigid than the second layer of material (45), wherein the first layer of material (50) is attached at the surface (44, best seen Figure 3-3) of the second layer of material (45). The construction of the modified Ovzinsky provides for either of the inferior tube portion or the superior tube portion of the gas delivery tubes to apply the rigidizing element (50) - "varying thickness along its length, e.g., to vary the stiffness or rigidity of the tube 42 along its length" (Para 0131); however, the explicit disclosure is to the superior tube portion (42.2, best seen Figure 2-2) "at the top of the patient's head" (Para 0131) to have a thicker construction - which equates to more stiffness or rigidity; whilst, the inferior tube portion (42.1, best seen Figure 2-1) "at the patient's cheeks" (Para 0131) to have a thinner construction - which equates to less stiffness or rigidity.

As to Claim 8, the modified Ovzinsky, specifically Veliss teaches the use of a "rigidizing element 50 may have a varying thickness along its length, e.g., to vary the stiffness or rigidity of the tube 42 along its length" (Para 0131), wherein the thicker regions provide "more shape-holding, self-supporting, and/or robust" support (Para 0133); whilst, the thinner regions "may be adjusted for comfort" (Para 0133). The construction of the modified Ovzinsky provides for either of the inferior tube portion or the superior tube portion of the gas delivery tubes to apply the rigidizing element (50) - "varying thickness along its length, e.g., to vary the stiffness or rigidity of the tube 42 along its length" (Para 0131); however, the explicit disclosure is to the superior tube portion (42.2, best seen Figure 2-2) "at the top of the patient's head" (Para 0131) to have a thicker construction - which equates to more stiffness or rigidity; whilst, the inferior

tube portion (42.1, best seen Figure 2-1) "at the patient's cheeks" (Para 0131) to have a thinner construction - which equates to less stiffness or rigidity.

As to Claim 9, the modified Ovzinsky, specifically Ovzinsky discloses the use of a seam ("The two conduit portions 621 and 622 can then be seam welded as shown in FIG. 13 (e.g., RF weld to couple the conduit portions) and then ultrasonically die cut as shown in FIG. 14 (e.g., to remove seam edges) to create an air tight textile conduit." Column 12, Lines 50-60) in the construction of the gas delivery tubes.

As to Claim 10, the modified Ovzinsky, specifically Ovzinsky discloses the construction wherein the gas delivery tube (x20) can be constructed with two parts that are joined together - "the air delivery conduit 620 includes first and second conduit portions 621, 622 that cooperate to form the conduit, i.e., first conduit portion provides a portion of the conduit circumference and the second conduit portion provides the remaining portion of the conduit circumference. In the illustrated example, the first and second conduit portions are symmetrical, however it should be appreciated that the conduit portions may be asymmetrical' (Column 12, Lines 30-40). Although the modified Ovzinsky **does not expressly disclose** the configuration wherein the inferior tube portion is joined to the superior tube portion, this configuration is obvious to try from a finite number of predictable results suitable for constructing an air impermeable tubes having the properties to vary the thickness/stiffness/rigidity along the length of the tube from the head region for stabilization to the patient contacting region near the seal forming structure for the purpose of providing patient comfort. (See Veliss Para 0131 and 0133). Consequently, it would have been obvious to one having ordinary skill in the art to select the manner of construction of the tube with respect to the desired characteristics of the tube since

it has been held where the general conditions of a claim are discussed in the prior art, discovering the optimum or workable ranges involves only routine skill in the art. Moreover, Applicant has not asserted the specific construction provides a particular advantage, solves a stated problem, or serves a purpose different from that of enabling the construction of multi-part tubes to formulate a tube having the desired operational characteristics of air impermeability and varied rigidity along the length of the tube; thus the use of the specific construction lacks criticality in its design. Consequently, one of ordinary skill in the art would have expected Applicant's invention to perform equally well with the modified Ovzinsky, as the construction methodology of the tube would yield the predictable results of providing a gas delivery tube that is air impermeable and suitable for adaptive rigidity along the length of the tube. Therefore, it would have been obvious to one having ordinary skill in the art to select the construction of the gas delivery tube with respect to the desired operational characteristics of air impermeability and varied rigidity along the length of the tube as taught by the modified Ovzinsky for the purpose of providing a suitable gas delivery tube for the positioning and stabilizing structure.

As to Claim 11, the modified Ovzinsky, specifically Ovzinsky discloses the use of various construction methodologies - "The film laminate 650 is applied to the fabric 655 (FIG. 11), and then the fabric and laminate are both thermoformed to create symmetrical complementary shapes (FIG. 12), i.e., conduit portions 621, 622. During the thermoforming process, the laminate adheres to the fabric giving it air impermeable properties. The two conduit portions 621 and 622 can then be seam welded as shown in FIG. 13 (e.g., RF weld to couple the conduit

portions) and then ultrasonically die cut as shown in FIG. 14 (e.g., to remove seam edges) to create an air tight textile conduit." (Column 12, Lines 45-65).

As to Claim 12, the modified Ovzinsky, specifically Ovzisky discloses the construction of the gas delivery tube (x20) having dual layers (best seen Figures 10-14) wherein "The film laminate 650 is applied to the fabric 655 (FIG. 11), and then the fabric and laminate are both thermoformed to create symmetrical complementary shapes (FIG. 12), i.e., conduit portions 621, 622." (Column 12, Lines 45-55). Consequently, as shown in Figures 10-14 the outer layer (655) is made of fabric (textile) material and the inner layer (650) is made of film laminate material.

As to Claims 13 and 14, the modified Ovzinsky, specifically Ovzisky discloses the construction of the gas delivery tube (x20) having dual layers (best seen Figures 10-14) wherein "The film laminate 650 is applied to the fabric 655 (FIG. 11), and then the fabric and laminate are both thermoformed to create symmetrical complementary shapes (FIG. 12), i.e., conduit portions 621, 622." (Column 12, Lines 45-55). As shown in Figures 10-14, both outer layers (655) are made of fabric (textile) material and both inner layers (650) is made of film laminate material.

As to Claim 15, the modified Ovzinsky, specifically Ovzinsky discloses the application of "spacer fabric" (Column 17, Lines 1 thru Column 19, Line 25) in the construction of the gas delivery conduit, wherein "the air delivery conduit 1220 includes a tube 1225 constructed of spacer fabrics and textile cover portions 1231, 1232 (e.g., laminated fabric) that substantially enclose the tube. Laminated fabric in cohesion with spacer fabrics provides a crush resistant textile conduit. Such conduit is crush resistant, air impermeable, form holding, and includes

possible in-built filtering qualities." (Column 17, Lines 1-10). Consequently, in this construction "The textile cover portions 1231, 1232 may be constructed of a laminated fabric (e.g., an inner layer of a film laminate 1250 (e.g., polyurethane or medical grade film) and an outer layer of a textile or fabric 1255 (e.g., synthetic or specified fabric)), each of which may be thermoformed to create its shape and then the textile cover portions may be seam welded to couple the textile cover portions" whereby "The tube 1225 is constructed of spacer fabrics including one or more outer layers 1225(1) (e.g., first and second layers) that cooperate to form the wall of the tube and an inner layer 1225(2) (e.g., third layer) supported within the internal passage provided by the one or more outer layers. The tube includes a relatively flat or non-cylindrical cross-section with the inner layer providing an anti-crush or occlusion resistant structure. The inner layer may define one or more passages or lumens through the tube." (Column 17, Lines 10-35). Thus, the intermediate layer of spacer fabric material was known.

As to Claims 16-21 and 23, each of the following claims recite the limitation whereby the inferior tube has a greater rigidity/thickness or imparts a rigidizing element than the superior tube. As previously addressed in several of the aforementioned claims, the modified Ovzinsky, specifically Veliss teaches the use of a "rigidizing element 50 may have a varying thickness along its length, e.g., to vary the stiffness or rigidity of the tube 42 along its length" (Para 0131), wherein the thicker regions provide "more shape-holding, self-supporting, and/or robust" support (Para 0133); whilst, the thinner regions "may be adjusted for comfort" (Para 0133). Although Veliss provides the explicit example for the rigidizing element applied to the superior tube - there is no structure that precludes, prevents, or hinders, the orientation wherein this rigidizing element is applied to the inferior tube portion instead.

Regarding Claim 16, as the modified Ovzinsky discloses the "varying thickness along its length, e.g., to vary the stiffness or rigidity of the tube 42 along its length" (Para 0131), the inferior tube portion may have a greater rigidity to the superior tube portion.

Regarding Claim 17, as the modified Ovzinsky discloses the "varying thickness along its length, e.g., to vary the stiffness or rigidity of the tube 42 along its length" (Para 0131), the inferior tube portion may have a greater wall thickness to the superior tube portion.

Regarding Claim 18, as the modified Ovzinsky discloses the "varying thickness along its length, e.g., to vary the stiffness or rigidity of the tube 42 along its length" (Para 0131), the inferior tube portion may have at least one additional layer in the form of the rigidizing element (50) as applied to the surface (44, best seen Figure 3-3) thus imparting a greater rigidity than that of the superior tube portion.

Regarding Claim 19, as the modified Ovzinsky discloses the "varying thickness along its length, e.g., to vary the stiffness or rigidity of the tube 42 along its length" (Para 0131), the inferior tube portion may have more stiffness and thus be less extensible than the materials of the superior tube portion.

Regarding Claim 20, as the modified Ovzinsky discloses the "varying thickness along its length, e.g., to vary the stiffness or rigidity of the tube 42 along its length" (Para 0131), the inferior tube portion may have the rigidizing element (50) as applied to the surface (44, best seen Figure 3-3) thus imparting a greater rigidity than that of the superior tube portion.

Regarding Claim 21, as the modified Ovzinsky discloses the "varying thickness along its length, e.g., to vary the stiffness or rigidity of the tube 42 along its length" (Para 0131), the inferior tube portion may have at least one additional layer in the form of the rigidizing element

(50) as applied to the surface (44, best seen Figure 3-3) this construction provides a greater number of layers and thus imparts a greater rigidity than that of the superior tube portion.

Regarding Claim 23 as the modified Ovzinsky discloses the "varying thickness along its length, e.g., to vary the stiffness or rigidity of the tube 42 along its length" (Para 0131), the inferior tube portion has a greater rigidity than the superior tube portion.

As to Claim 25, the modified Ovzinsky, specifically Veliss teaches the use of a "rigidizing element 50 may have a varying thickness along its length, e.g., to vary the stiffness or rigidity of the tube 42 along its length" (Para 0131), wherein the thicker regions provide "more shape-holding, self-supporting, and/or robust" support (Para 0133); whilst, the thinner regions "may be adjusted for comfort" (Para 0133). Veliss teaches the application of the first layer of material (50) as more rigid than the second layer of material (45), wherein the first layer of material (50) is attached at the surface (44, best seen Figure 3-3) of the second layer of material (45). The construction of the modified Ovzinsky provides for either of the inferior tube portion or the superior tube portion of the gas delivery tubes to apply the rigidizing element (50) - "varying thickness along its length, e.g., to vary the stiffness or rigidity of the tube 42 along its length" (Para 0131); however, the explicit disclosure is to the superior tube portion (42.2, best seen Figure 2-2) "at the top of the patient's head" (Para 0131) to have a thicker construction - which equates to more stiffness or rigidity; whilst, the inferior tube portion (42.1, best seen Figure 2-1) "at the patient's cheeks" (Para 0131) to have a thinner construction - which equates to less stiffness or rigidity. Thus the superior tube portion has a greater rigidity than the inferior tube portion.

As to Claim 24, , the modified Ovzinsky, specifically Veliss teaches the use of a "rigidizing element 50 may have a varying thickness along its length, e.g., to vary the stiffness or rigidity of the tube 42 along its length" (Para 0131), wherein the thicker regions provide "more shape-holding, self-supporting, and/or robust" support (Para 0133); whilst, the thinner regions "may be adjusted for comfort" (Para 0133). Veliss teaches the application of the first layer of material (50) as more rigid than the second layer of material (45), wherein the first layer of material (50) is attached at the surface (44, best seen Figure 3-3) of the second layer of material (45). The construction of the modified Ovzinsky provides for either of the inferior tube portion or the superior tube portion of the gas delivery tubes to apply the rigidizing element (50) - "varying thickness along its length, e.g., to vary the stiffness or rigidity of the tube 42 along its length" (Para 0131); however, the explicit disclosure is to the superior tube portion (42.2, best seen Figure 2-2) "at the top of the patient's head" (Para 0131) to have a thicker construction - which equates to more stiffness or rigidity; whilst, the inferior tube portion (42.1, best seen Figure 2-1) "at the patient's cheeks" (Para 0131) to have a thinner construction - which equates to less stiffness or rigidity. Thus the superior tube portion has a greater resistance to bending by the application of a thicker region.

5. Claims 22 and 26 are rejected under 35 U.S.C. 103 as being unpatentable over Ovzinsky et al. (9,731,090) in view of Veliss et al. (2008/0060649), as applied to Claim 16 and further in view of Scheiner et al. (20150352306).

As to Claim 22, please see the rejection of Claim 16. As previously addressed in several of the aforementioned claims, the modified Ovzinsky, specifically Veliss teaches the use of a "rigidizing element 50 may have a varying thickness along its length, e.g., to vary the stiffness or rigidity of the tube 42 along its length" (Para 0131), wherein the thicker regions provide "more shape-holding, self-supporting, and/or robust" support (Para 0133); whilst, the thinner regions "may be adjusted for comfort" (Para 0133). Although Veliss provides the explicit example for the rigidizing element applied to the superior tube - there is no structure that precludes, prevents, or hinders, the orientation wherein this rigidizing element is applied to the inferior tube portion instead. Thus, the differences in rigidity between inferior tube portion and the superior tube portion resulting a greater rigidity of the inferior tube portion as compared to the superior tube portion was known. Yet, **does not expressly disclose** this variation based on the "different" woven structure of the textile layer.

Scheiner teaches the construction methodology of interlocking fabrics imparts variations in the stretch and flexibility of fabrics. In particular, "Knitted fabrics may have different stretchability characteristics compared to woven fabrics. Knitted fabrics are typically more flexible than woven fabrics, which may only stretch in one direction (depending on the yarn they are made from), and therefore may provide a more comfortable fit for the patient. Knitted textiles may be constructed in such a way that the fabric has a two-way stretch--i.e. a first yarn oriented in a first direction has a lower flexibility than a yarn oriented in a second direction." (Para 0585). In light of this variation in woven structure through the use of different interlocking techniques, the weaved fabric has less stretch directions - "only stretch in one direction" than that of knitted fabrics which has "a two-way stretch", wherein the resultant

effect is that knitted fabrics "provide a more comfortable fit for the patient". Therefore, it would have been obvious to one having ordinary skill in the art to select the desired interlocking weave structure of the fabric with respect to the ability of the fabric to readily conform the user/patient during use. In this particular construction whereby the interior tube portion is more rigid than the exterior tube portion, the interior tube portion may be constructed from a weaving interlocking technique whilst the exterior tube portion may be constructed of a knitting interlocking technique, thus imparting a greater rigidity to the interior tube portion and greater comfort to the user/patient at the exterior tube portion region. This configuration is obvious to try from a finite number of predictable results having a recognized benefit of imparting variation/variability of rigidity along the gas delivery tube.

Therefore, it would have been obvious to one having ordinary skill in the art to modify the gas delivery tube of the modified Ovzinsky to have an interior tube portion constructed of a different woven structure than that of the exterior tube portion as taught by Scheiner to impart variations in the stretch and flexibility of fabrics.

As to Claim 26, the modified Ovzinsky, specifically Ovzisky discloses the positioning and stabilizing structure as shown in Claim 1 having a seal forming structure (510 - "the patient interface 510 (e.g., nasal prong/nozzle arrangement)" Column 7, Lines 30-35) in a therapeutically effective position on a patient's head for positive pressure airway treatment of sleeping disordered breathing ("The present technology relates to air delivery conduits used in Positive Airway Pressure (PAP) systems for treatment, e.g., of Sleep Disordered Breathing (SDB)" Column 1, Lines 10-20). Regarding the pressure characteristics, "In use, the PAP device generates a supply of pressurized air (e.g., 2-30 cm H.sub.2O) that is delivered to the patient

interface via the air delivery conduit. The patient interface or mask may have suitable configurations as is known in the art, e.g., full-face mask, nasal mask, oro-nasal mask, mouth mask, nasal prongs, nozzles, cradle, etc. Also, headgear may be utilized to comfortably support the patient interface in a desired position on the patient's face." (Column 6, Lines 5-10).

Yet, **does not expressly disclose** the explicit pressure being "at least 6 cmH20 above ambient air pressure throughout the patient's respiratory cycle in use... for positive pressure airway treatment of sleep disordered breathing.".

Scheiner teaches "the patient interface is configured to maintain a therapy pressure in a range of about 4 cmH2O to about 30 cmH2O, e.g., typically about 10 cmH2O, above ambient air pressure in use, throughout the patient's respiratory cycle, while the patient is sleeping, to ameliorate sleep disordered breathing, e.g., sleep apnea" (Para 0054) is a known pressure value suitable for "ameliorate sleep disordered breathing". As the value of 10 cm of H2O is greater than and also "at least 6 cm of H2O" this pressure is explicitly known to be suitable for "ameliorate sleep disordered breathing". Therefore, it would have been obvious to one having ordinary skill in the art to modify the pressure values of the positioning and stabilizing structure of the modified Ovzinsky to operate at "at least 6 cm of H2O" as taught by Scheiner for "ameliorate sleep disordered breathing".

Conclusion

6. Any inquiry concerning this communication or earlier communications from the examiner should be directed to ANNETTE F DIXON whose telephone number is (571)272-3392. The examiner can normally be reached M-F 9-5 EST with flexible hours.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Kendra D Carter can be reached on 571-272-9034. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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ANNETTE FREDRICKA DIXON
Primary Examiner
Art Unit 3782

/Annette Dixon/
Primary Examiner, Art Unit 3785